Adaptability Beats Added Structure: A Learnable Per-Head Positional Bias, Selected by Elimination, and What Its Length Extrapolation Could Mean for the Energy Footprint of Long-Context Inference

(applied note — exploratory, seeking review)

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Abstract

We report a small, deliberately falsifiable applied-ML result and discuss its forward-looking implications without overclaiming. Starting from a geometric prior (a positional-attention bias derived from a Calabi–Yau–type period), we ran a sequence of from-scratch training comparisons on WikiText-2 and were led, by a chain of negative and null results, to a simple conclusion: among a family of increasingly expressive per-head positional biases, the one that wins at GPU scale is the *plainest*—a single *learnable* linear slope per attention head ("learnable-ALiBi"), with no log-curvature term, no number-theoretic sequence, no per-head content scale, and no per-head softmax temperature. Three independent head-to-head studies (an attribution experiment, a six-family bake-off, and a content/position-balance probe) converge on this scheme; every added degree of freedom we tested either ties it within noise or *regresses* once the model is properly trained and regularized. The scheme PASSes a pre-committed quality gate (+8.1% perplexity over the best baseline on a regularized 8000-step run) and—the property we find most interesting—*extrapolates flat* in sequence length, degrading gracefully out to 8× the training context where learned absolute positions collapse. We are explicit that this is a known-good idea (ALiBi with learnable slopes), modestly improved and carefully validated, not a new mechanism. We then discuss, as *illustrative order-of-magnitude reasoning and not as a measured result*, why robust train-short / serve-long extrapolation is precisely the kind of property that bears on the energy and CO_2 cost of LLM serving at scale, and we lay out a concrete, fair-comparison roadmap (vs. NoPE, RoPE + position-interpolation, and learnable-ALiBi on a genuine long-range benchmark) that would be needed to substantiate any such claim. Code, raw runs, and an open benchmark dataset accompany this note.

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1 Scope, and what we do *not* claim

This is an applied side-note to a mathematics preprint [1], not a core result. To keep the reader's expectations calibrated, we state our limits up front.

- The mechanism we end on—**a learnable per-head linear positional bias**—is *not new*. It is ALiBi [2] with the per-head slopes made trainable rather than fixed. Our contribution is empirical and methodological: a controlled elimination showing that, on this benchmark, nothing more elaborate survives proper training, together with an honest measurement of its length-extrapolation behaviour.
- All quality numbers are on **WikiText-2** [6], a single small corpus, with models trained *from scratch* at the 25M-parameter scale. They are suggestive, not conclusive; they should be reproduced at larger scale and on more corpora before being relied upon.
- We claim **no memory or FLOP advantage** over a modern kernel. A learnable per-head scalar bias is $\mathcal{O}(1)$ state and $\mathcal{O}(L)$ to apply—identical to ALiBi as implemented inside FlashAttention [4]. Earlier framings of this project that suggested a memory "breakthrough" were against a materialized-table strawman and are retracted.
- The energy / CO_2 discussion in Section 5 is **explicitly illustrative**. We did not measure datacenter power. We reason about where a robust-extrapolation positional scheme *could* reduce work, and we are careful to separate that reasoning from evidence.

The reason we publish at all is the discipline: a bold prior was falsified, a false positive was caught, a modest true result was confirmed, and a controlled experiment then corrected our own story about *why* it works. We think that arc is worth recording even though the surviving mechanism is humble.

2 Background: positional bias as an additive score

In a decoder attention layer with head h, query i and key $j \le i$, the pre-softmax score is

$$s_h(i,j) \;=\; \frac{q_i^\top k_j}{\sqrt{D}} \;+\; \beta_h(i,j), \qquad j \le i, \tag{1}$$

where β_h is an additive positional bias and D the head dimension. Several widely used schemes are special cases of (1):

$$\begin{aligned} \text{learned-absolute:}\quad & \beta_h(i,j) = 0,\ \text{positions added to embeddings;} && (2)\\ \text{ALiBi [2]:}\quad & \beta_h(i,j) = -a_h\,(i-j),\ a_h \text{ a \textit{fixed} per-head slope;} && (3)\\ \text{NoPE [3]:}\quad & \beta_h(i,j) = 0,\ \text{no positional signal at all.} && (4) \end{aligned}$$

ALiBi's fixed geometric slope ladder $a_h = 2^{-8h/H}$ extrapolates well but is not adapted to the data; learned-absolute fits well in-distribution but cannot represent positions beyond the training length, so it collapses on extrapolation. The question that organizes this note is whether a *more expressive but still cheap* β_h can keep ALiBi's extrapolation while improving in-distribution quality—and, if so, *how much* expressiveness actually helps.

3 Method and protocol

The candidate family. We consider per-head biases of increasing expressiveness, all $\mathcal{O}(1)$ in stored parameters per head:

$$\text{(slope only)} \quad \beta_h(d) = -\,a_h\,d, \qquad a_h \text{ learnable}; \tag{5}$$

$$\text{(slope + log)} \quad \beta_h(d) = -\,a_h\,d + b_h \log d, \qquad a_h, b_h \text{ learnable}; \tag{6}$$

$$\text{(content balance)} \quad s_h(i,j) = c_h \frac{q_i^\top k_j}{\sqrt{D}} - a_h\,d, \qquad a_h, c_h \text{ learnable}; \tag{7}$$

$$\text{(+ temperature)} \quad s_h(i,j) = \tau_h^{-1}\Big(c_h \frac{q_i^\top k_j}{\sqrt{D}} - a_h\,d\Big), \qquad a_h, c_h, \tau_h \text{ learnable}, \tag{8}$$

with $d = i - j$. Equation (5) is *learnable-ALiBi*, our eventual selection. Equation (6) adds a learnable log-curvature (this is also where a fixed Calabi–Yau curve, $\beta_h(d) = -\gamma_h \log S_{20}(d)$, sits as a constrained special case). Equation (7) introduces a per-head *content scale* c_h that lets a head trade off content matching against positional decay—making the "local head vs. global head" split explicit rather than emergent. Equation (8) adds a per-head softmax temperature.

Pre-committed gate. Following the ALiBi evaluation methodology—train one model *from scratch* per positional scheme at matched compute—we fixed, in advance, a quality gate: PASS within +1% of the best baseline validation perplexity, KILL at > 5% regression. For each more-expressive scheme we also pre-committed a *specific* null: e.g. "content balance must improve length extrapolation over learnable-ALiBi, or it is killed."

Training. Decoder-only Transformers, $d_{\text{model}} = 512$, 8 layers, 8 heads ($\approx$ 25M parameters), trained on real WikiText-2 (`Salesforce/wikitext`, `wikitext-2-raw-v1`) on a single NVIDIA L4, PyTorch 2.9.1 / CUDA 12.9. The decisive runs use 4000 steps, context 512, batch 24, with ℓ_2 regularization on the learnable slopes ($\lambda = 10^{-3}$) and validation early-stopping—both of which proved *essential* (see Section 6). Length extrapolation is measured as validation perplexity at $1\times, 2\times, 4\times, 8\times$ the training context, with no retraining.

4 Results

4.1 Attribution: the gain is learnability, not the geometric shape

A controlled nested experiment (6) separates the linear slope from the curvature term (Table 1). A plain learnable slope with *zero* curvature already beats a fixed geometric (Calabi–Yau) bias; and when the curvature b_h is left free starting from the geometric value, gradient descent drives it *away* from that value (toward and past zero). The geometry was a prior, and not a well-pointed one; what helps is letting the data choose the slope.

4.2 Scale check: the win is real and PASSes the gate

On a regularized 8000-step run, the learnable scheme reaches 3.324 validation perplexity versus 3.617 for the best baseline (learned-absolute)—a +8.1% **improvement that PASSes** the pre-committed gate, with the overfitting that plagued an earlier under-regularized run (Section 6) removed.

Table 1: Attribution (local controlled run). Perplexity vs. fixed ALiBi; lower ppl is better. A learnable slope with no curvature (*alibi_learn*) already beats the fixed geometric shape (*holo_fixed*).

scheme	ppl @512	vs. fixed ALiBi
nested (slope + free log, linear init)	8.548	+6.9%
nested (slope + free log, geometric init)	8.680	+5.5%
learnable-ALiBi (slope only)	8.835	+3.8%
fixed geometric (Calabi–Yau) shape	8.998	+2.0%
ALiBi (fixed slopes)	9.181	—

Table 2: Length extrapolation (NVIDIA L4, from scratch, ctx 512, 4000 steps). Validation perplexity at 1–8× the training context; lower is better. The added content scale ties learnable-ALiBi; the temperature knob regresses.

scheme	1× (512)	2× (1024)	4× (2048)	8× (4096)
ALiBi (fixed)	3.705	3.623	3.577	3.513
learnable-ALiBi	**3.651**	**3.559**	3.500	3.429
+ content scale c_h	3.642	3.548	**3.488**	**3.428**
+ content + temperature	3.732	3.659	3.599	3.548
NoPE (content only)	10.290	10.845	11.571	12.311

4.3 Elimination: nothing more expressive survives

The decisive comparison (Table 2) trains the schemes (5)–(8) from scratch and measures length extrapolation. The explicit content-balance knob c_h (7) only *ties* learnable-ALiBi (+0.35% at 4×, +0.17% at 8×—within noise), and adding the per-head temperature τ_h (8) *regresses* at every length. The content-only control (NoPE) collapses and, tellingly, gets *worse* with length, confirming the positional term is doing real work.

Observation 1. *Across all three studies, expressiveness beyond a single learnable per-head slope does not survive proper training at this scale: it either ties within noise or regresses. We summarize this as* adaptability beats added structure—up to one learnable slope per head, and no further*.*

4.4 The property worth keeping: graceful extrapolation

The learnable scheme not only edges the baselines in-distribution; its perplexity is *flat-to-improving* from 1× to 8× the training context (Table 2), whereas learned-absolute positions degrade sharply beyond the training length (in earlier runs, perplexity rose from 4.3 at 1× to 20.6 at 4×). This is the inherited ALiBi property, preserved when the slopes are made learnable. We regard *stable train-short / serve-long extrapolation*, rather than the modest in-distribution gain, as the result's most consequential feature—and the bridge to the next section.

5 Forward-looking discussion: extrapolation and the energy footprint of serving

A caveat, restated. Nothing in this section is measured. We did not profile datacenter power, and we make no quantitative energy claim. What follows is order-of-magnitude reasoning about *where* a robust-extrapolation positional scheme could reduce computational work, offered to motivate the roadmap in Section 7, not to assert a result.

Why extrapolation, specifically, touches energy. The dominant energy costs in the LLM lifecycle that a positional scheme can plausibly influence are: (i) *long-context adaptation*—the additional training/fine-tuning used to extend a model's usable context (e.g. position-interpolation or RoPE-scaling passes); and (ii) *serving headroom*—the practice of training or fine-tuning at a longer context than strictly necessary to guarantee quality at the longest expected prompt. A positional scheme that degrades gracefully beyond its training length attacks both: if a model trained at context L retains quality at 4–$8L$ without a dedicated extension pass, the energy of that pass is avoided, and the training context itself can be set by the *typical* rather than the *worst-case* prompt.

An illustrative budget (not a claim). Self-attention cost scales as $\mathcal{O}(L^2)$ per layer in compute and—inside FlashAttention—roughly $\mathcal{O}(L)$ in HBM traffic. A context-extension fine-tune is itself a training job whose energy is a non-trivial fraction of a serving fleet's monthly budget at scale. *If* graceful extrapolation removed the need for such a pass for a class of deployments, the saving would be the entire energy of that pass; *if* it allowed a fleet to train at L rather than $2L$, the quadratic term in the affected training compute would fall by up to $4\times$ on those steps. We deliberately phrase these as conditionals: each depends on the extrapolation holding at production scale and quality, which our 25M-parameter WikiText-2 evidence does *not* establish.

What is genuinely free here. Independently of the above, the scheme adds essentially zero inference overhead: H learnable scalars, applied as an additive bias already fused into FlashAttention's inner loop for ALiBi. So whatever extrapolation benefit exists comes at no measurable per-token energy cost—unlike methods that buy context length with extra computation. This is the honest, defensible half of the energy story: *the mechanism is cheap*; the *open question* is whether its extrapolation translates into avoided training/fine-tuning at scale.

Datacenter and CO_2 framing. Inference now dominates the lifecycle energy of widely deployed models, and context length is a primary driver of per-request cost. A positional scheme that is (a) free at inference and (b) robust to serving beyond its training length is, in principle, aligned with reducing both per-request energy variance and the amortized training energy attributable to long-context support. Translating "aligned in principle" into "kWh and CO_2 saved" requires the measurements we have not made. We flag this as the single most valuable thing a reader with a serving fleet could contribute.

6 Lessons learned (methodological)

1. **A short training budget crowns the overfitter.** An early 1200-step screen ranked the most expressive bias first; at 6000 steps the ranking *inverted* (the expressive bias overfit a 2 MB corpus over $\sim$37 epochs, train loss $3\times$ lower but validation $3\times$ worse). Screens must validate at, or near, the target budget—or include explicit generalization controls.
2. **Expressive positional biases need a generalization budget, not a fit budget.** ℓ_2 regularization on the learnable slopes plus validation early-stopping converted the inversion into a clean +8.1% PASS. Without them the same scheme fails.
3. **Attribution beats celebration.** Once a learnable bias beat the baselines, the decisive experiment was not to ship it but to ask *why* it won. The nested control showed the win was learnability, not the geometric prior we started from—and saved us from publishing a false mechanism.

4. **Test the orthogonal axis, then stop.** Having exhausted "more distance shape," we tested the one untried axis (content/position balance). It tied. Knowing *when the search is done*—when added structure no longer pays—is itself a result.
5. **Negative and null results are the product.** Three of our four studies returned "no improvement." Reported together, they are what gives the surviving positive result its credibility.

7 Vision, roadmap, and open invitations

Vision. We are interested in positional mechanisms that are *cheap, adaptable, and robust to serving beyond their training regime*, because that triple is where a small architectural choice can have an outsized, fleet-level energy footprint. Learnable-ALiBi is a minimal instance; the forward question is how far the "adaptability beats added structure" principle generalizes (to RoPE frequencies made learnable, to per-layer rather than per-head slopes, to mixtures that remain $\mathcal{O}(1)$ state).

Roadmap (concrete, falsifiable).

1. **Validate on a real open-weight model.** The cheapest *valid* test is not from-scratch but a surgical edit of an ALiBi-native open-weight model (frozen RoPE models cannot be tested by a positional swap—that is the invalid frozen-swap that produces a vacuous PASS). Concretely: take BLOOM-1b1 (or MPT-7B at scale), make its fixed per-head slopes trainable (initialized at the released ladder), and run a short, parameter-efficient continued-pretrain; evaluate on a train-short/serve-long battery (RULER, PG-19 sliding-window perplexity, passkey retrieval) against the frozen fixed-ALiBi release. The primary metric is the *usable-context multiple per unit of adaptation compute*, which directly tests item 2. A full specification (models, licenses, intervention, pre-committed kill criteria) is in `docs/BENCHMARK_SPEC_OPENWEIGHT.md`.
2. **Fair long-range bake-off (from scratch, family-neutral).** Learnable-ALiBi vs. NoPE vs. RoPE + position-interpolation vs. fixed ALiBi at $\geq 10\times$ the parameter scale used here, pre-committing the gate and the extrapolation protocol—the cleaner but costlier complement to the open-weight test above.
3. **Measure the energy claim or retract it.** Instrument a serving harness; quantify, in kWh and CO_2e, the cost of a position-interpolation extension pass that graceful extrapolation could avoid. Report a measured number or state that no saving was found.
4. **Learnable RoPE frequencies.** Apply the same attribution discipline to RoPE: are learnable per-head frequency scales the RoPE analogue of our result, and do extra degrees of freedom again fail to survive?
5. **Exact-kernel orthogonal track.** A separate finding (rescaled-integer $QK^\top$ is $\sim 34\times$ more accurate than FP16 at L=2048, bottlenecked by the FP16 softmax) suggests an exact-$QK^\top$ + FP32-softmax kernel; pair it with learnable-ALiBi and re-measure quality and latency.

Invitations. We would especially value: (a) a refutation or confirmation of the extrapolation behaviour at larger scale; (b) a fair comparison against the baselines above run by someone without our priors; and (c) a measured energy/CO_2 accounting from anyone operating a serving fleet. Code, raw run logs, and an open benchmark dataset are linked from the repository [1]; the full chronological arc, including the false positive we caught, is documented there as a case study in falsification-first applied research.

Reproducibility

All runs use the public scripts in `4_ai_hardware_attention/` of the repository; the decisive comparison is `cy_sieve_hetero_pos.py`, raw outputs are in `gpu_phase_runs/hetero_pos_gpu_20260623.{js` and the open benchmark dataset (callensxavier/cy-sieve-attention-benchmark) carries the per-scheme perplexities and the documented arc.

References

[1] X. Callens, *Mirror-Map-Sieve* (code, raw runs, and the companion preprint v7), `https://github.com/xaviercallens/Mirror-Map-Sieve`, 2026.

[2] O. Press, N. A. Smith, M. Lewis, *Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation* (ALiBi), ICLR 2022.

[3] A. Kazemnejad et al., *The Impact of Positional Encoding on Length Generalization in Transformers* (incl. NoPE), NeurIPS 2023.

[4] T. Dao et al., *FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness*, NeurIPS 2022.

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[6] S. Merity, C. Xiong, J. Bradbury, R. Socher, *Pointer Sentinel Mixture Models* (WikiText), 2016.